

SPORTS HALL BOOKING SYSTEM

Business Objectives & System Configuration Documentation

Project Name	Sports Hall Booking System	Platform	ServiceNow Service Portal
Document Type	Business Objectives & Spec	Phase	Phase 1 - Analysis & Planning
Date	29 June 2026	Status	In Progress

1. Project Overview

The Sports Hall Booking System is a comprehensive digital solution developed on ServiceNow's Service Portal platform. It aims to completely streamline and automate the end-to-end process of reserving sports hall facilities. By replacing manual, paper-based, or fragmented booking methods with an efficient, central, and user-friendly online portal, the system maximizes utilization and optimizes operations.

2. Problem Statement

Current challenges faced by the organization without this centralized system include:

- Manual booking processes frequently leading to scheduling conflicts, errors, and double bookings.
- Absence of centralized real-time visibility regarding sports hall availability.
- Difficulty in managing, prioritizing, and processing multiple simultaneous user requests.
- Lack of automated tracking, instant notifications, and confirmations for reservations.
- Administrative staff spending excessive hours managing schedules and handling booking disputes manually.
- No reliable tracking or reporting mechanisms to calculate facility utilization patterns and peak usage times.

3. Business Objectives

3.1 Primary Objectives

- Digitize and automate the sports hall reservation process from end to end.
- Completely eliminate double-bookings and scheduling conflicts through dynamic, real-time availability checks.
- Reduce total administrative booking workload by at least 60% through structured automation workflows.

- Establish a single, robust centralized portal for all booking, approval, and management activities.
- Ensure stable 24/7 self-service system access for all authorized internal and external stakeholders.

3.2 Operational Objectives

- Enable users to effortlessly view, book, modify, and cancel their sports hall reservations completely online.
- Implement automated backend approval routing logical workflows based on predefined criteria.
- Trigger instant, dynamic automated email notifications and confirmations to users and system administrators.
- Maintain a full, secure digital audit trail of all historical bookings, updates, and cancellations.
- Empower facility administrators to block specific dates/times instantly for maintenance, emergency shutdowns, or special events.

3.3 User Experience Objectives

- Deliver a highly intuitive, modern, and mobile-responsive portal interface utilizing the ServiceNow Service Portal framework.
- Minimize transaction friction, lowering the average booking completion time to under 3 minutes per reservation.
- Enhance self-service tools to significantly reduce direct user dependency on facility administrative support.
- Provide users with a dedicated personal view to track upcoming bookings and access historical data.

3.4 Reporting & Analytics Objectives

- Generate granular facility utilization reports demonstrating peak usage times, patterns, and popular time slots.
- Track long-term reservation trends to support informed strategic resource planning and future capacity expansion.
- Provide real-time interactive dashboard visibility for administrative users summarizing daily and weekly bookings.

4. Success Criteria

The project deployment will be considered fully successful when the following metric thresholds are achieved:

- 100% of sports hall bookings are executed entirely through the online Service Portal without manual intervention.
- Zero occurrences of double-bookings or scheduling conflicts after the official system go-live date.
- User adoption rate achieves or surpasses 80% within the first 30 days post-launch.
- System availability and platform uptime are maintained strictly at 99% or above.
- Facility administrator booking management overhead time is lowered by at least 60%.

- All booking applicants receive structured automated confirmation emails within 5 minutes of form submission.

5. Key Stakeholders

Stakeholder	Role	Core Interest
Students / Staff	End Users	Frictionless, easy booking access and real-time slot views.
Facility Manager	Admin / Approver	Effective slot management, operations tracking, and rapid approvals.
IT Department	System Owner	Robust system integration, security compliance, and direct support.
Management	Sponsor	Comprehensive operational reporting, trend metrics, and cost efficiency.

6. Scope Matrix

6.1 In Scope

- Online self-service sports hall booking interface via ServiceNow Service Portal.
- Dynamic real-time availability calendar widget and clear time slot selection.
- Automated lifecycle emails and confirmation alerts (Event-driven notifications).
- Pre-configured multi-tier booking approval engine for administrators.
- Personalized user dashboards containing booking history and self-service cancellation capabilities.
- Administrative analytics dashboards for checking real-time schedules and running usage reports.

6.2 Out of Scope

- Payment processing gateways or immediate financial fee collection mechanisms (Deferred to Phase 2).
- Native mobile application development (iOS/Android native apps).
- Direct synchronization with third-party external calendar applications like Google Calendar or Outlook (Deferred to Phase 2).
- Equipment inventory rental management modules.

7. High-Level Timeline

Project Phase	Target Activities	Current Status
Phase 1	Requirement Analysis & Planning	In Progress
Phase 2	System Design & Architecture	Pending
Phase 3	UI/UX Development & ServiceNow Customization	In Progress
Phase 4	Testing & Rigorous Quality Assurance	Pending
Phase 5	Deployment, Training & Go-Live	Pending

8. ServiceNow Technical Implementation: Events & Notifications

This technical guide defines the configuration rules for the Event Registry and Automated Email Engine inside ServiceNow to drive automated reminder updates.

Activity 1: Open Event Registry

Navigate to the Application Navigator. Search for Event Registry. Under the System Notification module, select **Event Registry**.

Activity 2: Create a New Event Record

Click the **New** action button to spin up a fresh Event Registry configuration sheet.

Activity 3: Configure Event Structural Properties

Fill the required fields using the explicit configuration metadata mapping listed below, then click Submit.

Event Registry Property	Target Configuration Value
Event Name	sports.hall.reminder
Table	u_sports_hall_bookings
Priority	100 (Default System Priority)
Fired By	Scheduled Jobs Script Engine
Description	System event triggered automatically when a designated Scheduled Job evaluates booking parameters.
Application Scope	Global

8.1 Email Notification Configuration

To configure the outbound email layer, navigate to **Application Navigator** → **System Notification** → **Email** → **Notifications**. Create a new record matching these properties:

Notification Property	Target Configuration Value
Name	Booking confirmation reminder
Table	u_sports_hall_bookings
When to Send	Event is fired
Event Name	sports.hall.reminder
Active	True (Checked)

Expected Operational Outcome: When the ServiceNow background Scheduled Job executes, it scans records and fires the `sports.hall.reminder` event. The platform's notification engine catches the event, automatically binding it to the *Booking confirmation reminder* rule, formatting the layout, and transmitting an automated notification to the targeted recipient.

9. ServiceNow Schema Architecture: Table & Fields Customization

Detailed technical structural definitions mapping the custom attributes created within the ServiceNow database layer to support system parameters.

Activity 1: Navigation — Open the standard Application Navigator, search for Tables, and head directly to **System Definition > Tables**.

Activity 2: Allocation — Select **New** to declare a custom table canvas.

Activity 3: Label Definition — Populate Label = Sports Hall Bookings and Name = `u_sports_hall_bookings`, then click Submit.

Activity 4: Column Generation — Enter the Table Column sub-tab layout and create the exact field specifications listed below.

Field Label	Field Database Name	Data Type	Max Length	Default Choice / Value
Refund Payment	u_refund_payment	Currency	20	<i>None</i>
Transaction Number	u_transaction_number	String	40	<i>None</i>
Reminder	u_reminder	True/False	--	False
Duration Hours	u_duration_hours	Decimal	20	<i>None</i>
Hall Name	u_hall_name	String	40	<i>None</i>
Instructions	u_instructions	String	1000	<i>None</i>
Status	u_status	Choice	40	Pending
Mode of Refund	u_mode_of_refund	String	40	<i>None</i>
End Time	u_end_time	Date/Time	--	<i>None</i>
Updated by	u_updated_by	String	40	<i>None</i>
Updates	u_updates	Integer	--	<i>None</i>
Created by	u_created_by	String	40	<i>None</i>
Payment	u_payment	Currency	20	<i>None</i>

9.1 Status Field Choice Values Matrix

The choice list options configured specifically for the u_status field choice dictionary must contain the following structural mapping values:

Choice Interface Label	Backend Database Choice Value
Pending	pending
Approved	approved
Rejected	rejected
Cancelled	cancelled
Completed	completed

10. Stakeholder Management & Mapping Strategy

This section outlines stakeholder placement, active operational responsibilities, and influence matrices utilized to manage communications and milestones during development.

Stakeholder	Role Context	Core Responsibilities	Influence	Interest
Project Sponsor	Provides project approval & funding	Reviews and signs off on project scope, financial budget, and production deployment permissions.	High	High
Sports Facility Administrator	Manages sports hall operations	Dynamically sets up slots, organizes user data, processes manual requests, and coordinates maintenance blocks.	High	High
Employees / Students	End users of the platform	Leverages portal to track time slot availability, issues system booking forms, and executes cancellations.	Low	High
Approver / Manager	Reviews formal booking requests	Audits individual escalated requests, authorizing or rejecting bookings via corporate policy workflows.	High	High
ServiceNow Developer	Develops and optimizes application	Handles full Service Portal UI design, configures internal logic flows, maps Flow Designer items, and tests scripts.	High	High
System Administrator	Maintains instances & platform	Oversees broad instance health, role assignments, ACL compliance, platform performance, and database instance backups.	Medium	Medium
Testing Team	Validates overall code quality	Conducts regression tests, functional validation, load stress testing, and structures User Acceptance Testing (UAT).	Medium	High
Support Team	Handles post-deployment care	Monitors logs post-launch, debugs visual components, manages routine updates, and acts as help desk support.	Medium	Medium

10.1 Power / Interest Engagement Matrix

Based on individual quadrant classification, stakeholders are mapped into specific managerial strategic tracks:

Stakeholder Entity	Power Level	Interest Level	Target Management Strategy
Project Sponsor	High	High	Manage Closely
Sports Facility Administrator	High	High	Manage Closely
Approver / Manager	High	High	Manage Closely
Employees / Students	Low	High	Keep Informed
ServiceNow Developer	High	High	Manage Closely
System Administrator	Medium	Medium	Keep Satisfied
Testing Team	Medium	High	Keep Informed
Support Team	Medium	Medium	Monitor Regularly

10.2 Structured Stakeholder Communication Plan

- **Project Sponsor:** Receives standard automated high-level project status dashboards, timeline reports, and crucial phase milestone sign-offs.
- **Sports Facility Administrator:** Actively participates in routine operational requirement reviews, initial sprint testing cycles, and live cutover reviews.
- **Approver / Manager:** Consulted specifically during the structural mapping of the workflow routing nodes and policy logic criteria setup.
- **Employees / Students:** Targeted group for feedback collection loops during late User Acceptance Testing (UAT) cycles and immediate post-go-live surveys.
- **Development Team:** Engages in daily agile technical alignment touchpoints to track development progress, clear blockers, and structure object commits.
- **Support Team:** Kept updated via detailed technical transition manuals, early service architecture reviews, and post-implementation monitoring cadence.

Sports Hall Booking System | Business Objectives & Technical Specifications Guide | Skill Wallet Framework